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Neer classification of proximal humeral fractures

Last revised by Joachim Feger (/users/jfeger?lang=us) on 17 Nov 2025

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The **Neer classification of proximal humeral fractures** utilizes specific terminology that influences the classification, which is essential to include in radiology reports of proximal humeral fractures.

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Usage

The Neer classification is the most frequently used system alongside the AO classification of proximal humeral fractures (/articles/aoota-classification-of-proximal-humeral-fractures?lang=us) (c. 2025) [10].

Fracture parts

The Neer system divides the proximal humerus (/articles/humerus?lang=us) into four parts and considers not the fracture line, but the displacement as being significant for the classification.

The four parts are:

1. humeral head
2. greater tubercle
3. lesser tubercle
4. humeral shaft

Displacement

Displacement is on a per-part basis. A fractured part is considered displaced if angulation exceeds 45°, or if the fracture is displaced by more than 1 cm [1].

The simplest displaced fracture, which is possible, therefore, is a two-part fracture, i.e. two distinct parts of the humerus. A minimally displaced fracture, even one that includes multiple fracture lines, however, merely constitutes a type I, one-part fracture as the humerus appears as a single entity.

Classification

The Neer classification has been variably adapted by multiple authors. The classification consists of four major groupings, based on the number of displaced parts. Neer added a fifth group [9].

One-part fracture

- fracture lines involve 1 to 4 parts
- none of the parts are displaced (i.e. <1 cm and <45°)

These non-displaced or minimally displaced fractures account for ~70-80% of all proximal humeral fractures and are almost always treated conservatively [6,7].

Two-part fracture

- fracture lines involve 2 to 4 parts
- one part is displaced (i.e. >1 cm or >45°)

Four possible types of two-part fractures exist (one for each part):

1. surgical neck (/articles/humerus?lang=us): most common
2. greater tubercle
   - frequently seen in the setting of anterior shoulder dislocation [8]
   - a lower threshold of displacement (>5 mm) has been proposed [8]
3. anatomical neck

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4. lesser tubercle: uncommon

These fractures account for approximately 20% of proximal humeral fractures [6].

Three-part fracture

- fracture lines involve 3 or 4 parts
- two parts are displaced (i.e. >1 cm or >45°)

Two three-part fracture patterns are encountered [7]:

1. greater tubercle and shaft are displaced with respect to the lesser tubercle and articular surface, which remain together
   - most common three-part pattern
2. lesser tubercle and shaft are displaced with respect to the greater tubercle and articular surface, which remain together

These fractures account for approximately 5% of proximal humeral fractures [6].

Four-part fracture

- fracture lines involve all 4 parts
- three parts are displaced (i.e. >1 cm or >45°) with respect to the 4th

These fractures are uncommon (<1% of proximal humeral fractures) [6].

This pattern has poor non-operative results, and as the articular surface is no longer attached to any parts of the humerus which are attached to soft tissues. This pattern has a high incidence of osteonecrosis (/articles/osteonecrosis-2?lang=us) [7].

These fractures require operative management.

Valgus impacted four-part fracture

This type of four-part fracture is considered its own category and has a different prognosis than typical four-part fractures [9].

History and etymology

The Neer classification was created by **Charles S Neer II**, an American orthopedic surgeon and professor. Dr Neer founded and was the first President of the American Shoulder and Elbow Surgeons.

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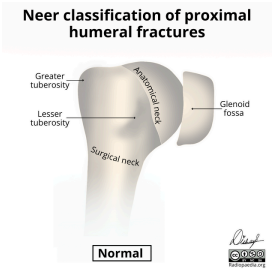

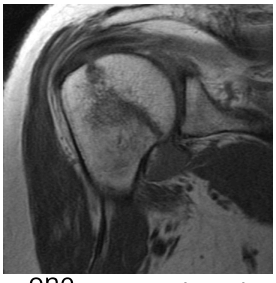

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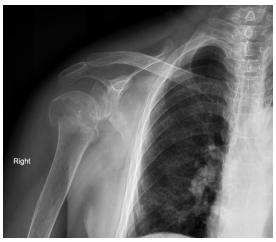

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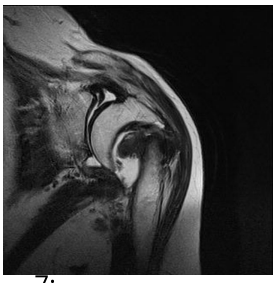

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